Forecasting U.S. Inflation Using Time Series and Machine Learning Models: The Role of Unemployment

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Abstract

This report examines whether unemployment improves one-month-ahead forecasts of U.S. inflation. The analysis uses monthly U.S. macroeconomic data from 1990 to 2026, with inflation constructed as year-over-year CPI inflation. The forecasting models include a naive benchmark, ARIMA/SARIMA, ARIMAX/SARIMAX, VAR, Ridge Regression, and XGBoost. The main empirical comparison is between inflation-only models and models that include unemployment together with monetary policy and real activity controls. The results show that SARIMA provides the best out-of-sample performance over the 2019–2026 testing period and across pre-COVID, COVID-inflation surge, and post-COVID sub-periods. In contrast, adding unemployment does not consistently improve forecast accuracy. ARIMAX and SARIMAX underperform their non-exogenous counterparts, Ridge performs worse when unemployment is included, and XGBoost gains only a negligible improvement. VAR Granger causality tests also do not provide strong evidence that unemployment helps predict inflation in this setup. Overall, the evidence suggests that short-term U.S. inflation forecasts are mainly driven by inflation persistence and seasonal time series dynamics rather than labor market information.

Keywords: inflation forecasting; unemployment; Phillips Curve; ARIMA; SARIMA; VAR; Ridge Regression; XGBoost; machine learning.

Contents

1 Introduction

Inflation forecasting is an important task in macroeconomics because inflation affects household purchasing power, firms' pricing decisions, financial markets, and monetary policy. For central banks, accurate inflation forecasts are especially important because monetary policy affects the economy with a delay. However, inflation is difficult to forecast because it depends on many factors, including past inflation, labor market conditions, monetary policy, real economic activity, and unexpected shocks.

This report examines whether unemployment helps improve forecasts of U.S. inflation. The motivation comes from the Phillips Curve, which suggests that labor market conditions may be related to inflation dynamics (1). When unemployment is low, the labor market may become tighter, wage pressure may increase, and inflationary pressure may rise. When unemployment is high, weaker labor market conditions may reduce inflationary pressure. Therefore, unemployment is a natural candidate predictor for inflation.

This report does not aim to estimate or prove the Phillips Curve as a structural macroeconomic theory. Instead, the Phillips Curve is used as a theoretical motivation for an empirical forecasting exercise. The main question is whether unemployment improves inflation forecast accuracy. This follows a broader debate in the inflation forecasting literature: Stock and Watson (2) show that real activity variables can be useful for forecasting inflation, while Atkeson and Ohanian (3) argue that simple benchmark forecasts are often difficult to beat.

To answer this question, the report compares traditional time series models and machine learning models using monthly U.S. macroeconomic data. Inflation is constructed from the Consumer Price Index, while unemployment is the main predictor of interest. Additional variables such as the federal funds rate and industrial production growth are included to capture monetary policy and real economic activity.

The models used in this report include a naive forecast, ARIMA/SARIMA, ARIMAX/SARIMAX, VAR, Ridge Regression, and XGBoost. Forecast performance is evaluated using out-of-sample error metrics, mainly Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The key comparison is between models that use only past inflation and models that also include unemployment.

The report focuses on four research questions:

RQ1. How well do traditional time series models forecast U.S. inflation?

RQ2. Does adding unemployment improve inflation forecast accuracy?

RQ3. Do machine learning models outperform traditional time series models?

RQ4. Does unemployment contain dynamic predictive information for inflation in a VAR framework?

2 Motivation and Forecasting Framework

2.1 Phillips Curve as Motivation

The Phillips Curve provides the main reason for including unemployment in this report. It suggests that unemployment may be related to inflation through labor market conditions (1). A lower unemployment rate may indicate a tighter labor market, stronger wage pressure, and higher inflationary pressure. A higher unemployment rate may indicate weaker demand and lower inflationary pressure.

This report uses the Phillips Curve only as motivation. The purpose is not to test whether the Phillips Curve is always valid, but to examine whether unemployment is useful for forecasting inflation. If unemployment contains useful information about future inflation, then models that include unemployment should produce lower forecast errors than models that exclude it.

Recent applied work also supports the relevance of testing this relationship empirically. Qin (4) use a VAR framework to study the relationship between unemployment and inflation in the U.S. economy and apply Granger causality, impulse response functions, and variance decomposition. Their results suggest that unemployment can help improve inflation forecasts over their sample. This motivates the inclusion of unemployment in this report, while leaving its usefulness to be evaluated by out-of-sample forecast performance.

2.2 Traditional Time Series Models

The first group of models consists of traditional time series methods. These models are important because inflation often displays persistence, meaning that past inflation may contain useful information about future inflation. The naive forecast is used as the simplest benchmark. Although simple, it is important because a complex model is useful only if it can outperform a simple alternative.

ARIMA and SARIMA models forecast inflation using only its own past values and past forecast errors. They serve as inflation-only benchmarks and are standard tools in time series forecasting (5). SARIMA extends ARIMA by allowing for seasonal patterns, which may be relevant when working with monthly data. ARIMAX and SARIMAX extend these models by adding external predictors such as unemployment, the federal funds rate, and industrial production growth. This model group is central to the report because it directly tests the role of unemployment.

The VAR model is used to capture dynamic relationships among inflation, unemployment, interest rates, and industrial production growth. Unlike ARIMAX, which treats unemployment as an external predictor, VAR allows all variables in the system to interact dynamically (6). It also allows Granger causality tests to examine whether past unemployment helps predict inflation.

Recent comparative studies continue to use ARIMA and SARIMA as traditional benchmarks while comparing them with modern forecasting models such as XGBoost, LSTM, and TCN for inflation prediction (7). Other CPI forecasting research also includes ARIMA/SARIMAX-type models when studying U.S. economic trends (8).

2.3 Machine Learning Models

The second group of models consists of machine learning methods. These models are included because they may capture patterns that traditional linear time series models miss. Ridge Regression is used as a regularized linear model. It is useful when the model includes many lagged predictors because regularization reduces overfitting (9). In this report, Ridge Regression serves as a machine learning baseline using lagged inflation, unemployment, and other macroeconomic variables.

XGBoost is used as a nonlinear tree-based model. It can capture nonlinear relationships and interactions between variables (10). This is useful because the relationship between unemployment and inflation may not be purely linear or stable over time.

Recent CPI forecasting studies show that machine learning models can be useful in this setting. Gur (11) apply LSTM, MARS, XGBoost, LSTM-MARS, and LSTM-XGBoost to U.S. CPI forecasting using macroeconomic indicators such as oil prices, the federal funds rate, GDP, unemployment, and house prices. Henderi and Sofiana (7) compare traditional and modern time series models for inflation prediction and find that modern models such as LSTM and TCN can perform well in their dataset. However, the usefulness of machine learning depends on the sample, variables, and forecasting design.

To avoid data leakage, all machine learning models use only lagged predictors. The data are split chronologically rather than randomly, so the models are evaluated on future observations that were not used during training. The main logic is simple: if unemployment is useful for inflation forecasting, then models with unemployment should have lower MAE and RMSE than models without unemployment.

3 Data

3.1 Data Source and Sample Period

This report uses monthly U.S. macroeconomic data from the Federal Reserve Economic Data (FRED) database. Four variables are included: the Consumer Price Index for All Urban Consumers, the unemployment rate, the effective federal funds rate, and the industrial production index (12–15). These variables represent the main economic dimensions of the forecasting problem: inflation, labor market conditions, monetary policy, and real economic activity.

The raw sample covers January 1990 to April 2026, giving 436 monthly observations before transformations. This period is chosen because it provides a modern sample with enough observations for both traditional time series models and moderate-complexity machine learning models. It also covers several important macroeconomic episodes, including the Global Financial Crisis, the zero lower bound period, the COVID-19 shock, the post-2021 inflation surge, and the recent high-interest-rate environment.

After constructing year-over-year inflation and industrial production growth, the transformed sample begins in January 1991 and ends in April 2026, giving 424 observations. The empirical analysis uses a chronological train-test split shown in Table 1.

Table 1: Train-test split

Set	Period	Observations	Purpose
Training set	January 1991–December 2018	336	Model estimation and tuning
Testing set	January 2019–April 2026	88	Out-of-sample forecast evaluation

A chronological split is used because the data are time series. Random splitting is avoided because it would break the time ordering of the observations and could lead to data leakage.

3.2 Variables and Transformations

The target variable is U.S. inflation, constructed from the Consumer Price Index. The main predictor of interest is the unemployment rate. The federal funds rate and industrial production growth are included as additional macroeconomic predictors. Table 2 summarizes the variables.

Table 2: Variables used in the analysis

Variable	Code	Description	Role
Consumer Price Index	CPIAUCSL	Consumer Price Index for All Urban Consumers	Used to construct inflation
Unemployment Rate	UNRATE	Civilian unemployment rate	Main predictor of interest
Federal Funds Rate	FEDFUNDS	Effective federal funds rate	Monetary policy variable
Industrial Production Index	INDPRO	Index of real industrial production	Real activity variable

Inflation is constructed as year-over-year log inflation:

$$\pi_t = 100 \times [\log(CPI_t) - \log(CPI_{t-12})] . \tag{1}$$

Industrial production growth is also constructed as a year-over-year log growth rate:

$$IPG_t = 100 \times \left[\log(INDPRO_t) - \log(INDPRO_{t-12})\right]. \tag{2}$$

The unemployment rate and the federal funds rate are kept in levels. For forecasting models, lagged values of the variables are created so that only information available before the forecast date is used.

3.3 Data Preprocessing

The preprocessing procedure consists of several steps. First, all four series are converted to monthly date format and merged by observation date. Second, the sample is restricted to January 1990 through April 2026. Third, CPI is transformed into year-over-year inflation, and industrial production is transformed into year-over-year growth. Fourth, lagged variables are created for the forecasting models.

For the machine learning models, lagged features are created using 1-month to 12-month lags of inflation, unemployment, the federal funds rate, and industrial production growth. This produces 48 lagged predictors:

$$\begin{aligned} \mathcal{X}_t^{ML} = \{&\pi_{t-1}, \dots, \pi_{t-12}, \\ &UNRATE_{t-1}, \dots, UNRATE_{t-12}, \\ &FEDFUNDS_{t-1}, \dots, FEDFUNDS_{t-12}, \\ &IPG_{t-1}, \dots, IPG_{t-12}\}. \end{aligned} \tag{3}$$

After constructing 12-month lagged features, the effective machine learning dataset contains approximately 412 observations. This sample size is sufficient for moderate-complexity models such as Ridge Regression and XGBoost, but it is not large enough for data-intensive deep learning models.

3.4 Exploratory Data Analysis

The exploratory data analysis provides a preliminary view of the behavior of the variables before model estimation. It is used to describe the data, identify major macroeconomic episodes, and motivate the use of dynamic forecasting models. Table 3 reports descriptive statistics for the transformed variables from January 1991 to April 2026.

Table 3: Descriptive statistics

Variable	Count	Mean	Std. Dev.	Min	Median	Max
Inflation	424	2.582	1.488	-1.978	2.560	8.599
Unemployment Rate	424	5.663	1.761	3.400	5.300	14.800
Federal Funds Rate	424	2.739	2.192	0.050	2.455	6.910
Industrial Production Growth	424	1.380	4.256	-19.017	2.234	15.317

The average year-over-year inflation rate is about 2.58 percent, but inflation varies substantially over time. The minimum inflation rate is -1.98 percent in July 2009, reflecting deflationary pressure during the Global Financial Crisis. The maximum inflation rate is 8.60 percent in June 2022, corresponding to the post-pandemic inflation surge.

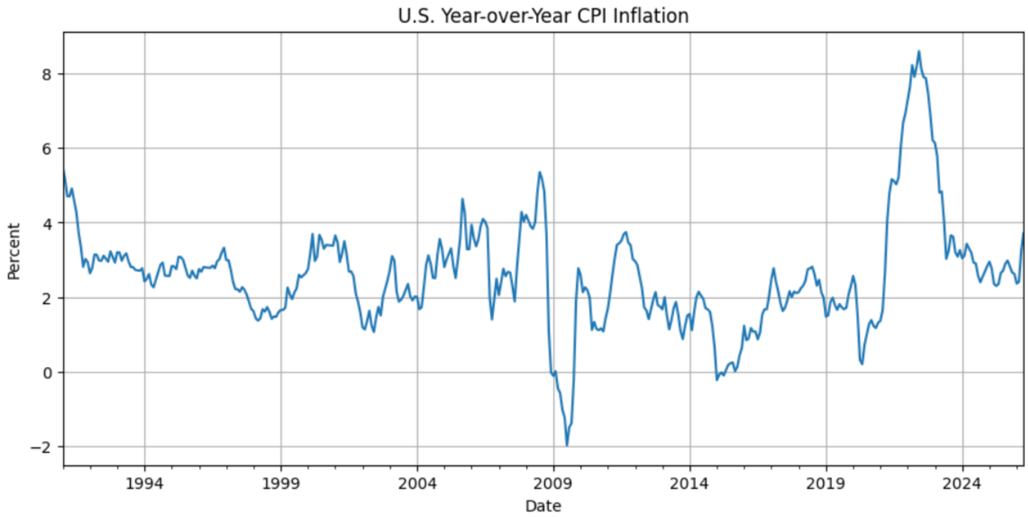
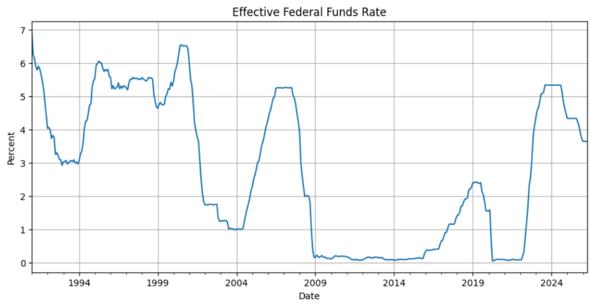


Figure 1: U.S. year-over-year CPI inflation

Figure 1 shows that inflation was relatively stable for much of the sample but became highly volatile during crisis periods. Inflation declined sharply during the Global Financial Crisis and increased rapidly after 2021. Figure 2 reports the remaining macroeconomic variables.

(a) Unemployment rate

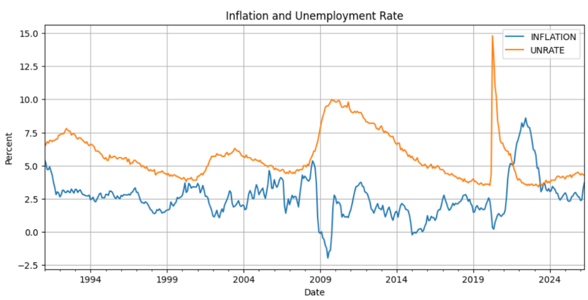


(b) Effective federal funds rate

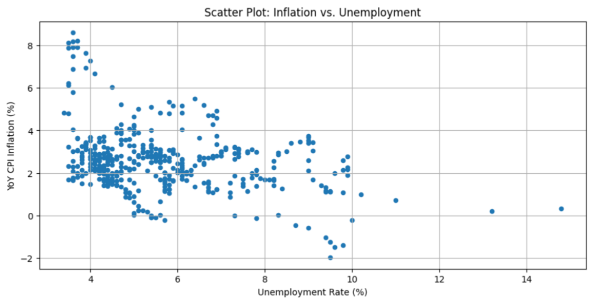
(c) Industrial production growth

Figure 2: Main macroeconomic predictors

The unemployment rate increases sharply during recessions, especially during the Global Financial Crisis and the COVID-19 shock. The federal funds rate reflects major changes in U.S. monetary policy, including the zero lower bound after the Global Financial Crisis and COVID-19, and the tightening cycle after the post-2021 inflation surge. Industrial production growth is more volatile, with a large decline in 2020 and rebound in 2021.



(a) Inflation and unemployment



(b) Inflation vs. unemployment

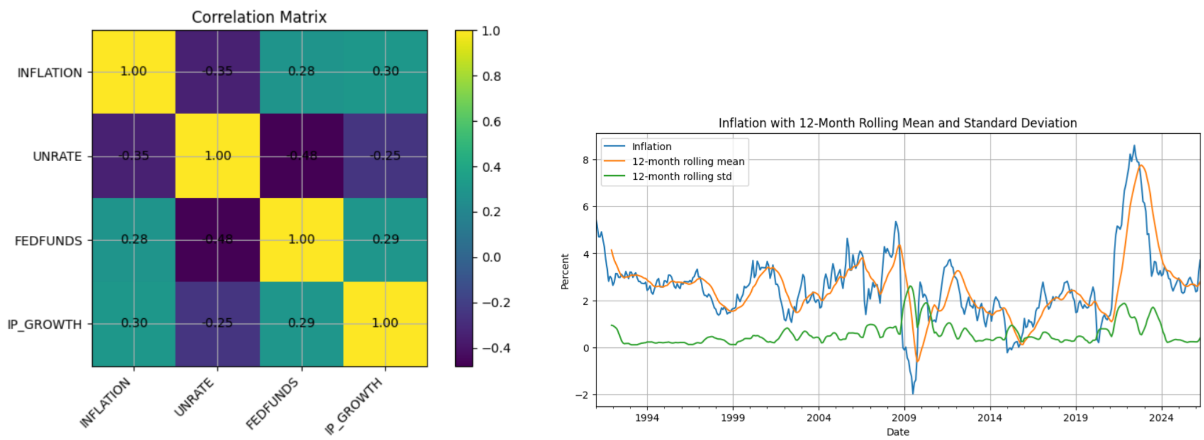
Figure 3: Inflation and unemployment relationship

Figure 3 is useful for visualizing the Phillips Curve motivation. In some periods, inflation and unemployment appear to move in opposite directions, but the relationship is not stable throughout the sample. The COVID-19 period shows a sharp increase in unemployment followed by a later inflation surge, which suggests that unemployment should be tested empirically rather than assumed to have a stable relationship with inflation.

Table 4: Correlation matrix

Variable	Inflation	UNRATE	FEDFUNDS	IP Growth
Inflation	1.00	-0.35	0.28	0.30
UNRATE	-0.35	1.00	-0.48	-0.25
FEDFUNDS	0.28	-0.48	1.00	0.29
IP Growth	0.30	-0.25	0.29	1.00

Inflation has a moderate negative correlation with unemployment, approximately -0.35. This is consistent with the basic Phillips Curve intuition, but it should not be interpreted as causal evidence. Because macroeconomic relationships may occur with lags, contemporaneous correlations are not sufficient to determine forecasting usefulness.



(a) Correlation matrix

(b) Inflation rolling mean and standard deviation

Figure 4: Descriptive diagnostics

Figure 4 shows that the post-2021 period stands out because both the level and volatility of inflation increased. This reinforces the need to evaluate forecasting models on a challenging test period that includes recent inflation instability.

4 Methodology

4.1 Forecasting Setup

The main objective is to evaluate whether unemployment improves the accuracy of U.S. inflation forecasts. The target variable is year-over-year CPI inflation, denoted by π_t. The forecasting task is one-month-ahead prediction:

$$\hat{\pi}_{t+1}, \tag{4}$$

where $\hat{\pi}_{t+1}$ is the predicted inflation rate for the next month.

The models are estimated using the training period and evaluated on the testing period. The forecasting exercise compares traditional time series models and machine learning models. The main comparison is between models that rely only on past inflation and models that also include unemployment and other macroeconomic variables.

4.2 Preliminary Time Series Diagnostics

Before estimating the models, several time series diagnostics are conducted. Stationarity is examined using the Augmented Dickey-Fuller test (16) and the KPSS test (17). The ADF test has the null hypothesis that the series contains a unit root, while the KPSS test has the null hypothesis that the series is stationary. Using both tests provides a more balanced view of the stationarity properties of the variables.

Autocorrelation and partial autocorrelation plots are used to inspect the dynamic structure of inflation. For ARIMA-family models, residual diagnostics are performed using the Ljung-Box test (18). For the VAR model, stability is checked using the roots of the estimated system.

4.3 Naive Forecast

The naive forecast is the simplest benchmark model. It assumes that inflation next month will be equal to inflation in the current month:

$$\hat{\pi}_{t+1} = \pi_t. \tag{5}$$

This benchmark is important because inflation is often persistent. A more complex model is useful only if it produces lower forecast errors than this naive alternative.

4.4 ARIMA and SARIMA Models

ARIMA models are used as traditional univariate time series benchmarks. They forecast inflation using only past values of inflation and past forecast errors. The general model is written as $ARIMA(p, d, q)$, where p is the autoregressive order, d is the differencing order, and q is the moving average order (5).

Because the data are monthly, SARIMA is also considered as a seasonal extension of ARIMA:

$$SARIMA(p, d, q)(P, D, Q)_s, \tag{6}$$

where P, D, and Q are seasonal autoregressive, differencing, and moving average orders, and $s = 12$ for monthly data. Since the target variable is year-over-year inflation, much of the seasonality is already reduced, but SARIMA is still included to test whether remaining annual dependence improves forecast accuracy.

4.5 ARIMAX and SARIMAX Models

ARIMAX extends ARIMA by adding external predictors. In this report, ARIMAX directly tests whether unemployment improves inflation forecasting. Conceptually,

$$\pi_t = f(\pi_{t-1}, \pi_{t-2}, \ldots, X_{t-1}, X_{t-2}, \ldots) + \epsilon_t, \tag{7}$$

where $X_t = [UNRATE_t, FEDFUNDS_t, IPG_t]$. The key comparison is between ARIMA and ARIMAX. If ARIMAX produces lower out-of-sample errors than ARIMA, this suggests that unemployment and other macroeconomic variables contain useful predictive information beyond past inflation.

SARIMAX is the seasonal version of ARIMAX. It allows both seasonal dynamics and external predictors. To avoid data leakage, exogenous variables are used in lagged form so that the model only uses information that would have been available at the forecast origin.

4.6 Vector Autoregression Model

The VAR model captures dynamic interactions among multiple macroeconomic variables (6). The VAR system is defined as

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + u_t, \tag{8}$$

where

$$Y_t = [\pi_t, UNRATE_t, FEDFUNDS_t, IPG_t]'. \tag{9}$$

The VAR model serves two purposes. First, it provides a multivariate time series forecast of inflation. Second, it allows Granger causality tests to examine whether past unemployment contains predictive information for inflation. The null hypothesis is

$$H_0: \text{ past unemployment does not help predict inflation.} \tag{10}$$

If the null hypothesis is rejected, this suggests that unemployment contains dynamic predictive information for inflation.

4.7 Ridge Regression

Ridge Regression is used as a regularized linear machine learning model. It is useful when the model contains many lagged predictors because regularization reduces overfitting by shrinking coefficients toward zero (9). The Ridge objective function is

$$\min_{\beta} \sum_{t=1}^{T} (\pi_t - \hat{\pi}_t)^2 + \lambda \sum_{j=1}^{k} \beta_j^2, \tag{11}$$

where λ controls the strength of regularization.

The Ridge model uses 12 lags of inflation, unemployment, the federal funds rate, and industrial production growth. This creates a feature set that captures inflation persistence, labor market conditions, monetary policy, and real economic activity.

4.8 XGBoost

XGBoost is included as a nonlinear machine learning model. It is based on gradient-boosted decision trees and can capture nonlinear relationships and interactions among predictors (10). In this report, XGBoost uses the same lagged feature set as Ridge Regression. This allows a fair comparison between a regularized linear model and a nonlinear tree-based model.

Because the dataset is small for machine learning, XGBoost is estimated with conservative hyperparameters, including limited tree depth, moderate learning rate, and regularization. The goal is not to maximize in-sample fit, but to improve out-of-sample forecast performance.

4.9 Forecast Evaluation Metrics

Forecast performance is evaluated using out-of-sample errors on the testing period. The two main metrics are MAE and RMSE:

$$MAE = \frac{1}{n} \sum_{t=1}^{n} |y_t - \hat{y}_t|, \tag{12}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^{n} (y_t - \hat{y}_t)^2}. \tag{13}$$

MAE measures the average absolute size of forecast errors. RMSE gives more weight to large errors and is especially useful when large forecast mistakes are costly. A model is considered better if it produces lower MAE and RMSE on the testing set.

4.10 Summary of Model Roles

Table 5: Summary of model roles

Model	Main purpose
Naive Forecast	Simple benchmark
ARIMA/SARIMA	Inflation-only time series benchmark
ARIMAX/SARIMAX	Tests whether unemployment and macro predictors improve forecasts
VAR	Captures dynamic relationships and tests Granger causality
Ridge Regression	Regularized linear machine learning baseline
XGBoost	Nonlinear machine learning model

5 Empirical Results

5.1 Time Series Diagnostics

Before comparing forecast performance, several diagnostics are conducted to examine the time series properties of the variables and the adequacy of the estimated models.

5.1.1 Stationarity Tests

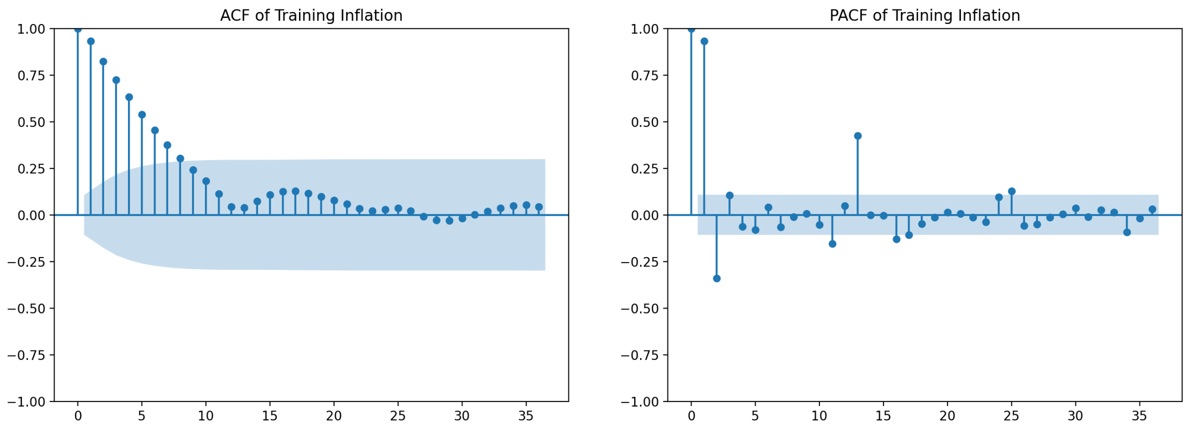
Table 6 reports the ADF and KPSS results for the main variables.

Table 6: Stationarity test results

Variable	ADF p-value	ADF conclusion	KPSS p-value	KPSS conclusion	Overall conclusion
Inflation	0.009	Reject unit root	0.010	Reject stationarity	Mixed evidence
Unemployment Rate	0.117	Fail to reject unit root	0.100	Fail to reject stationarity	Mixed evidence
Federal Funds Rate	0.272	Fail to reject unit root	0.010	Reject stationarity	Likely non-stationary
Industrial Production Growth	0.032	Reject unit root	0.100	Fail to reject stationarity	Likely stationary

The results show mixed evidence for inflation and unemployment. Inflation rejects the unit root null under the ADF test, but the KPSS test also rejects stationarity, suggesting that year-over-year inflation remains highly persistent. The unemployment rate also shows mixed evidence, while the federal funds rate appears more clearly non-stationary. Industrial production growth is the most stationary variable.

5.1.2 ACF and PACF of Inflation



(a) ACF of training inflation

(b) PACF of training inflation

Figure 5: Autocorrelation diagnostics for training inflation

The ACF plot shows strong inflation persistence: autocorrelations are high at short lags and decline gradually. The PACF plot shows a strong spike at lag 1 and a negative spike at lag 2. It also shows a visible spike around lag 13, suggesting that some annual structure remains even after using year-over-year inflation. This supports the inclusion of SARIMA and SARIMAX in the model comparison.

5.1.3 ARIMA Order Selection

The ARIMA order is selected using AIC over a restricted grid of candidate models. The lowest AIC is obtained by ARIMA(3,1,3), so this specification is selected as the main non-seasonal ARIMA model. The same non-seasonal order is used for ARIMAX to make the comparison between ARIMA and ARIMAX more direct.

Table 7: Top ARIMA specifications by AIC

Rank	ARIMA order	AIC
1	(3,1,3)	213.033
2	(3,0,2)	227.839
3	(1,0,3)	231.939
4	(2,0,3)	232.269
5	(1,0,2)	233.765

5.1.4 Residual Diagnostics and VAR Stability

The Ljung-Box test examines whether residuals from ARIMA-family models still contain autocorrelation. Table 8 shows that ARIMA and ARIMAX still have residual autocorrelation at lag 12, while SARIMA and SARIMAX do not. This suggests that the seasonal models capture the remaining annual dependence in inflation more effectively.

Table 8: Ljung-Box residual diagnostics

Model	Lag	p-value	Conclusion
ARIMA	6	0.410	No residual autocorrelation
ARIMA	12	0.033	Residual autocorrelation remains
ARIMA	24	0.190	No residual autocorrelation
ARIMAX	6	0.337	No residual autocorrelation
ARIMAX	12	0.025	Residual autocorrelation remains
ARIMAX	24	0.151	No residual autocorrelation
SARIMA	6	0.293	No residual autocorrelation
SARIMA	12	0.520	No residual autocorrelation
SARIMA	24	0.949	No residual autocorrelation
SARIMAX	6	0.115	No residual autocorrelation
SARIMAX	12	0.301	No residual autocorrelation
SARIMAX	24	0.835	No residual autocorrelation

The selected VAR lag order is 2, and the estimated VAR system is stable. The minimum root modulus is 1.013 and the maximum root modulus is 4.963, indicating that the VAR model is suitable for forecasting and Granger causality analysis in this setting.

5.2 Main Forecast Performance

Table 9 reports the out-of-sample forecast performance over the full test period from January 2019 to April 2026.

Table 9: Forecast performance over full test period, 2019–2026

Rank	Model	MAE	RMSE	Observations
1	SARIMA	0.225	0.306	88
2	SARIMAX	0.263	0.392	88
3	ARIMA	0.295	0.406	88
4	Naive	0.308	0.420	88
5	VAR	0.356	0.477	88
6	ARIMAX	0.420	0.535	88
7	Ridge	0.586	0.911	88
8	XGBoost	1.018	1.714	88

SARIMA is the best-performing model, with the lowest RMSE of 0.306. SARIMAX ranks second, followed by ARIMA and the naive forecast. This indicates that traditional time series models outperform the selected machine learning models in this forecasting exercise. The strong performance of SARIMA suggests that inflation's own historical dynamics and remaining seasonal structure are highly informative for one-month-ahead forecasting.

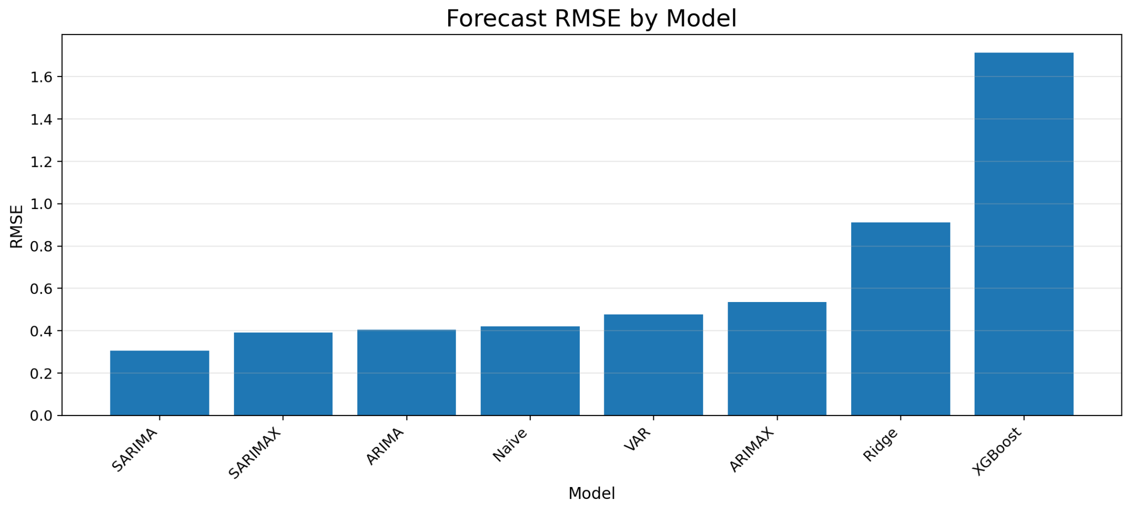


Figure 6: Forecast RMSE by model

5.3 Actual versus Forecasted Inflation

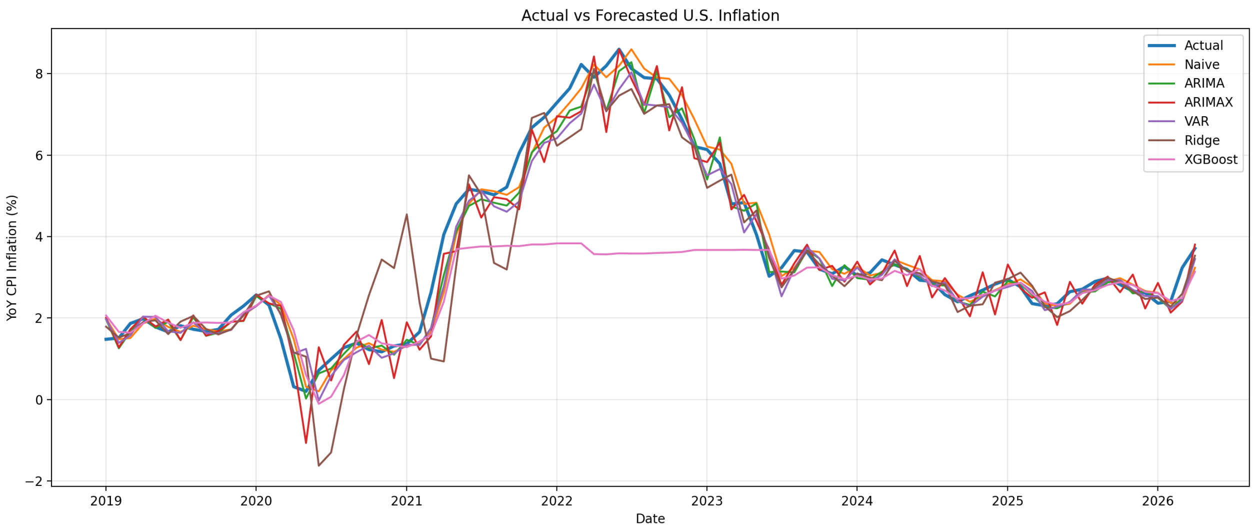


Figure 7: Actual versus forecasted U.S. inflation

Figure 7 shows that the main time series forecasts generally follow the movement of actual inflation. They capture the broad inflation increase during 2021–2022 and the decline after 2023. Ridge performs poorly during the early COVID period and around the inflation surge. XGBoost performs the worst because its forecasts remain relatively flat during the inflation surge, suggesting that it struggles to extrapolate beyond the training data range.

5.4 Does Unemployment Improve Inflation Forecasts?

Table 10 compares models with and without unemployment.

Table 10: Forecast comparison with and without unemployment

Comparison	RMSE without UNRATE	RMSE with UNRATE	Change in RMSE	Improves RMSE?
ARIMA vs ARIMAX	0.406	0.535	+0.129	No
SARIMA vs SARIMAX	0.306	0.392	+0.086	No
Ridge without UNRATE vs Ridge with UNRATE	0.623	0.911	+0.287	No
XGBoost without UNRATE vs XGBoost with UNRATE	1.720	1.714	-0.006	Yes, but very small

The results show that unemployment does not consistently improve inflation forecast accuracy. ARIMAX performs worse than ARIMA, SARIMAX performs worse than SARIMA, and Ridge also performs worse when unemployment is included. XGBoost improves slightly, but the improvement is extremely small. Therefore, unemployment does not provide robust additional predictive value for one-month-ahead U.S. inflation forecasting in this setup.

5.5 VAR and Granger Causality

The VAR model is also used to test whether unemployment contains dynamic predictive information for inflation. The Granger causality test evaluates the null hypothesis that unemployment does not Granger-cause inflation.

Table 11: VAR Granger causality result

Test	p-value	Conclusion at 5%	Selected VAR lag
UNRATE does not Granger-cause inflation	0.503	Fail to reject H_0	2

The p-value is 0.503, so the null hypothesis cannot be rejected. This means that past unemployment does not provide statistically significant predictive information for inflation in the VAR specification. This result is consistent with the forecast comparison: unemployment does not appear to be a strong predictor of short-term inflation in this sample.

5.6 Forecast Performance across Macroeconomic Regimes

Because the test period includes different macroeconomic regimes, forecast performance is evaluated across three sub-periods: pre-COVID, COVID and inflation surge, and post-COVID disinflation.

5.6.1 Pre-COVID Period: 2019

Table 12: Forecast performance during the pre-COVID period

Model	MAE	RMSE	Observations
SARIMA	0.135	0.176	12
SARIMAX	0.142	0.183	12
Ridge	0.171	0.205	12
VAR	0.189	0.232	12
Naive	0.205	0.235	12
XGBoost	0.206	0.244	12
ARIMA	0.216	0.258	12
ARIMAX	0.235	0.268	12

In 2019, forecast errors are relatively small across all models. SARIMA performs best, followed closely by SARIMAX. Since this window has only 12 observations, the ranking should be interpreted cautiously.

5.6.2 COVID and Inflation Surge Period: 2020–2022

Table 13: Forecast performance during the COVID and inflation surge period

Model	MAE	RMSE	Observations
SARIMA	0.331	0.420	36
SARIMAX	0.346	0.510	36
Naive	0.410	0.522	36
ARIMA	0.428	0.539	36
VAR	0.527	0.650	36
ARIMAX	0.583	0.717	36
Ridge	1.080	1.370	36
XGBoost	2.016	2.592	36

The 2020–2022 period is the most difficult forecasting window. Forecast errors increase sharply, especially for Ridge and XGBoost. SARIMA remains the best-performing model, followed by SARIMAX and Naive. The results suggest that the COVID shock and the post-COVID inflation surge created a major stress test for forecasting models. The fact that ARIMAX and SARIMAX underperform their non-exogenous counterparts also suggests that unemployment and other macroeconomic predictors do not help much during this stress period.

5.6.3 Post-COVID Disinflation Period: 2023–2026

Table 14: Forecast performance during the post-COVID disinflation period

Model	MAE	RMSE	Observations
SARIMA	0.147	0.195	40
ARIMA	0.198	0.286	40
SARIMAX	0.215	0.305	40
VAR	0.252	0.325	40
Ridge	0.266	0.350	40
Naive	0.252	0.355	40
ARIMAX	0.334	0.382	40
XGBoost	0.362	0.634	40

Forecast errors decline substantially after 2023. SARIMA again performs best, followed by ARIMA and SARIMAX. This indicates that once inflation becomes more stable, traditional time series models forecast inflation more accurately.

5.6.4 Summary of Sub-Period Results

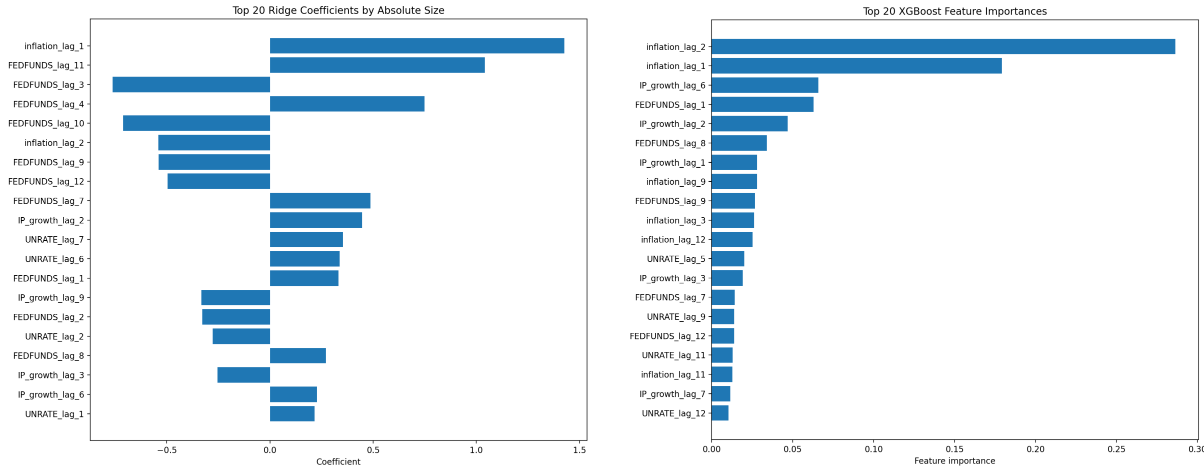
Table 15: Best model by evaluation window

Evaluation window	Best model	RMSE	Main interpretation
2019–2026	SARIMA	0.306	Best full-period performance
2019	SARIMA	0.176	Stable pre-COVID period
2020–2022	SARIMA	0.420	Best during COVID and inflation surge
2023–2026	SARIMA	0.195	Forecast accuracy improves after inflation stabilizes

SARIMA is the best model in all evaluation windows. This shows that the seasonal univariate time series model performs robustly across different macroeconomic regimes. Forecast errors are highest during 2020–2022 and decline after 2023, confirming that inflation forecasting becomes harder during structural breaks and easier when inflation stabilizes.

5.7 Feature Importance from Machine Learning Models

Although Ridge and XGBoost do not outperform the time series models, their feature importance results help identify which variables matter most in the machine learning models.



(a) Top Ridge coefficients by absolute size

(b) Top XGBoost feature importances

Figure 8: Machine learning feature importance

The Ridge coefficient plot shows that `inflation_lag_1` is the most important predictor. Several federal funds rate and unemployment lags also appear among the top features, but they are less dominant than lagged inflation. XGBoost feature importance also shows that lagged inflation dominates the forecast. The two most important variables are `inflation_lag_2` and `inflation_lag_1`. Unemployment lags appear in the top 20, but their importance is lower than inflation lags. This reinforces the conclusion that short-term inflation forecasts are mainly driven by inflation persistence rather than unemployment.

5.8 Summary of Empirical Findings

The empirical results show five main findings. First, SARIMA is the best-performing model over the full test period and across all sub-periods. Second, the naive forecast remains a strong benchmark, confirming that year-over-year inflation is highly persistent. Third, adding unemployment does not consistently improve forecast accuracy. Fourth, the VAR Granger causality test does not provide strong evidence that unemployment helps predict inflation. Fifth, forecast performance varies across macroeconomic regimes, with the largest errors during the COVID and inflation surge period.

Overall, unemployment is theoretically relevant, but it does not provide robust additional predictive value for one-month-ahead U.S. inflation forecasting in this empirical setup. Short-term inflation forecasts are mainly driven by inflation persistence and seasonal time series dynamics rather than labor market information.

6 Discussion

The empirical results show that traditional time series models outperform the selected machine learning models in this inflation forecasting exercise. SARIMA produces the lowest forecast errors over the full testing period and across all sub-periods, suggesting that inflation's own persistence and remaining seasonal structure are highly important for short-term forecasting.

This result is consistent with the ACF and PACF plots. Inflation shows strong autocorrelation, especially at recent lags, which explains why models based mainly on past inflation, such as Naive, ARIMA, and SARIMA, perform well. SARIMA performs best because it captures both short-run dynamics and remaining seasonal patterns.

The main research question is whether unemployment improves inflation forecasts. The results suggest that it does not improve forecasts consistently. ARIMAX performs worse than ARIMA, SARIMAX performs worse than SARIMA, and Ridge performs worse when unemployment is included. XGBoost improves only slightly, but the improvement is too small to change the main conclusion. The VAR Granger causality test also fails to find strong evidence that unemployment helps predict inflation.

This does not mean that unemployment is unrelated to inflation. Rather, in this dataset and one-month-ahead forecasting setup, unemployment does not provide strong additional predictive value beyond past inflation. The Phillips Curve remains a useful theoretical motivation, but the unemployment-inflation relationship appears unstable in this sample.

The sub-period results show that forecast performance depends on the macroeconomic regime. Errors are low in 2019, rise sharply during the COVID and inflation surge period, and decline again after 2023. The COVID period likely weakened the predictive usefulness of unemployment, since unemployment rose sharply in 2020 while inflation surged later in 2021–2022.

Finally, the machine learning results support the same interpretation. Ridge and XGBoost do not outperform the time series models, and feature importance results show that lagged inflation is more important than unemployment. Overall, short-term inflation forecasts in this report are mainly driven by inflation persistence rather than labor market information.

7 Limitations

This report has several limitations. First, the model uses a limited set of macroeconomic variables: inflation, unemployment, the federal funds rate, and industrial production growth. Other important inflation predictors, such as oil prices, energy prices, inflation expectations, housing costs, exchange rates, and supply chain indicators, are not included. These omitted variables may be especially important during the post-COVID inflation surge.

Second, the report focuses only on one-month-ahead forecasting. This is suitable for short-term prediction, but unemployment may have stronger predictive value over longer horizons. Therefore, the results should not be interpreted as a complete rejection of the Phillips Curve.

Third, the sample size is relatively small for machine learning models. Monthly macroeconomic data provide only a few hundred observations after transformations and lag construction. This may limit the performance of models such as XGBoost.

Fourth, the test period includes a major structural break caused by COVID-19 and the post-

COVID inflation surge. Models trained on pre-2019 data may not fully capture this unusual macroeconomic environment.

Fifth, the analysis uses revised historical data rather than real-time vintage data. In practice, forecasters use data available at the time, which may later be revised. Finally, this report focuses on forecasting accuracy rather than causal identification. The results show whether unemployment improves prediction, not whether unemployment causally affects inflation.

8 Conclusion

This report examines whether unemployment improves U.S. inflation forecasting using time series and machine learning models. The target variable is year-over-year CPI inflation, and the forecasting task is one-month-ahead prediction. The data cover the period from 1990 to 2026, with training data ending in December 2018 and testing data covering January 2019 to April 2026.

The main result is that SARIMA provides the best out-of-sample forecast performance. It outperforms Naive, ARIMA, ARIMAX, SARIMAX, VAR, Ridge Regression, and XGBoost over the full testing period. It also performs best in the pre-COVID, COVID and inflation surge, and post-COVID disinflation sub-periods. This suggests that inflation's own dynamics and remaining seasonal structure are highly informative for short-term forecasting.

The second main result is that unemployment does not consistently improve forecast accuracy. ARIMAX performs worse than ARIMA, SARIMAX performs worse than SARIMA, and Ridge performs worse when unemployment is included. XGBoost improves only slightly when unemployment is added. The VAR Granger causality test also does not provide strong evidence that past unemployment helps predict inflation.

Overall, the evidence suggests that short-term U.S. inflation forecasts in this setup are mainly driven by inflation persistence rather than unemployment. Although unemployment is theoretically motivated by the Phillips Curve, it does not provide robust additional predictive value for one-month-ahead inflation forecasting in this sample.

The sub-period analysis shows that forecast performance changes across macroeconomic regimes. Errors are highest during the COVID and inflation surge period, confirming that structural breaks make inflation forecasting more difficult. Future research could include oil prices, inflation expectations, housing costs, real-time data, longer forecast horizons, and more advanced machine learning or hybrid models.

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